

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7/1	(a)	(i)		Gene – a {section/ part/ piece/ sequence/ length} of {DNA/ chromosome} (that codes for a particular characteristic/ protein) Reject {strand/ molecule} of DNA	1			1		
		(ii)		Allele – (alternative/ different) {form/ version/ type/ variation} of a gene	1			1		
	(b)	(i)		Gametes Nn and Nn (1) Genotypes of offspring NN, Nn, Nn, nn (1) If candidates use different letters to N and n max 1.		2		2		
		(ii)		nn/ (homozygous) recessive (1) If had {dominant allele/ N} they would not {have cystic fibrosis/ be affected} /(1) Do not award second mark if genotype is given as NN or Nn or heterozygous or homozygous dominant		1	1	2		
		(iii)		{NN/ homozygous dominant} and {Nn/ heterozygous }		1		1		
	(c)			{Most phenotypes/ they} are {the result of/ controlled by/ have/ need} {multiple genes/ more than one (pair of) gene} / also affected by environment/ reference to epigenetics Accept more than one pair of genes	1			1		
				Question 7/1 total	3	4	1	8	0	0